CIT 253
Assignment 04 – MySQL and Beginning SQL

In this assignment, we will

1. first practice using the `mysql` client from the command-line, as shown in Chapter 4
2. practice SQL commands as shown in Chapter 5

Part 1: Setup tasks

To start, do the following steps:

1. Start up the MySQL server – Open the XAMPP Control Panel and click Start next to MySQL. This starts up the server.
2. Start up the `mysql` client – Open the command prompt and type `C:\xampp\mysql\bin\mysql -u root` to launch the client and connect to the server as the root user (who will manage all the other users).
3. Set a password for the root user – as shown on page 641 of the textbook Step 2, type the command
   ```
   SET PASSWORD FOR 'root'@'localhost' =
   PASSWORD('yourchosenpassword');
   ```
   Inside the parentheses, type whatever password you want, in place of `yourchosenpassword`.
4. You have now changed the password for the root user.
5. Now create another user, as shown on page 644, by typing the command `CREATE USER 'webuser'@'localhost' IDENTIFIED BY 'BroWs1ng';` (If you want to use a different username and password instead of `webuser` and `BroWs1ng`, then feel free to do so).
6. Make a screenshot of your command prompt showing that you have done the above steps. You will be submitting the screenshot as part of this assignment.
7. Type exit at the command prompt to exit the `mysql` client. Note that now the command prompt will still remain open – you have only exited the `mysql` client, not the command prompt itself.

To make running the `mysql` client easier, without having to specify the entire path to it at the command-line each time we want to run it as in the above steps, let us create a Windows batch file that will run the client for us. To do this, open Notepad, and

1. type the following two lines:
   ```
   set PATH=%PATH%;C:\xampp\mysql\bin
   mysql -u %1 -p
   ```
   [Explanation: The first line modifies the system environment Path variable by adding the location of the `mysql` client to it. The second line launches the `mysql` client. The `-u` and `-p` are called flags; whatever comes after `-u` is the username for whoever is trying to access the MySQL server via the `mysql` client; the `-p` indicates that we are going to use a password.]

2. Save as `runClient.bat` (remember to change the "Save as type:" to "All Files" rather than Text Documents).
   Once you have this batch file prepared, every time you want to run the `mysql` client, all you have to do is to open a command prompt, change to the folder where you have your `runClient.bat` file saved, and then type `runClient` followed by your username for the MySQL server.
3. Test your batch file by first starting up the `mysql` client and connecting to the server as the root user
   a. Open the command prompt
   b. Change to the folder where you saved the `runClient.bat` file
   c. Type `runClient root`
   d. You should now be prompted for the password. Enter the password that you set for the root user.
   e. Once you have successfully connected as root, make a screenshot of the command prompt.
   f. Type `exit` to exit the client. Leave the command prompt open.
      [An explanation of how this works at the command-line: the `%1` that you typed into the batch file in step #1 above is a placeholder – whatever you type after `runClient` at the command line will be plugged in, in place of `%1`, e.g., when you typed `runClient root` in step c) above, `root` was inserted in place of `%1`, making that line within the batch file work as if you had typed

      ```
      mysql -u root -p
      ```

      and you are therefore prompted to enter the password for `root`]
4. Try using your batch file to connect as the other user that you created.
   a. Type `runClient webuser` (if you used a different username, then type that in place of `webuser`)
   b. When prompted, type in the password.
   c. Once you have successfully connected, make a screenshot of the command prompt.
   d. Exit the `mysql` client

Part 2: Creating a database and tables, inserting data and practicing SQL queries

In this second half of the assignment, we will create a database, create tables in it, and add data to the tables, all using `SQL` commands via the `mysql` client, as shown in Chapter 5.

SQL commands that create tables within a database, insert data into those tables and so on can be rather long, and it can be easy to make mistakes when typing them in. For this reason, it is usually easier to type such SQL commands into a file, and then use that file with the `mysql` client. This allows you to correct errors in your SQL commands by simply editing that file, rather than having to retype the entire long command again and hope you don't make any errors. This kind of file, saved with a `.sql` file extension, is called an *SQL source file*. In this part of this

assignment, we will see how to prepare SQL source files, and then you will use them in ALL database tasks that you perform in this class.

1. Open Notepad.
2. Type the SQL code to create a database named `sitename` using the SQL command `CREATE` as shown on page 132 –

   ```
   CREATE DATABASE sitename;
   ```
3. Since the MySQL server could be hosting several databases as mentioned earlier, whenever we want to work within some specific database, we have to choose/set that database as the one currently being operated on. This is done with the USE command as shown on page 132. Type this line of code into the Notepad file below the one that creates the database.

   ```
   USE sitename;
   ```
4. Save as `createSitenameDatabase.sql` (remember to change the "Save as type:" to "All Files" rather than Text Documents – here and every time you create any `.sql` file in the future)
5. Close this file.
6. Open a new Notepad window and type the `SQL` code given on page 133, that creates the `users` table. Save as `createUsersTable.sql`
7. Close this file as well.
8. Now that we have prepared two SQL source files – one to create a database, and one to create a table within that database – we will use those files within the mysql client, to perform those two tasks.
9. Use your batch file to run the `mysql` client as `root` and enter your password when prompted.
10. Now we will start a logfile by using the `tee` command in MySQL. A logfile is a plain text file that captures everything that you did within MySQL. To start one up, at the command prompt in the `mysql` client, you would type a command in the format

    ```
    tee path to your logfile
    ```

    So, for this assignment, if you had a folder on your machine at the path `C:\CIT253\Assignment04`, for example, you would type

    ```
    tee C:\CIT253\Assignment04\chapter_5_log.txt
    ```

    and this would result in a file named `chapter_5_log.txt` being created in that folder, containing a log of all the commands that you type in the `mysql` client from here on, and their results. You will be submitting this logfile as part of this assignment.
11. At the `mysql` client's command line, type

    ```
    source createSitenameDatabase.sql;
    ```

    Once you press enter, either the database will be created successfully and the `mysql` command line's prompt will change to show sitename as the database currently being

used, or you will see error messages because you have errors in the SQL code within that file. If the latter, open that file in Notepad again and correct the errors.

12. Once the sitename database has been created successfully, at the `mysql` client's command line, type

    ```
    source createUsersTable.sql;
    ```

    Confirm that the table was created, as shown on page 134. If there were any errors in your SQL syntax within the file that you created above, correct those and try again.
13. Now that the `users` table has been created, we are ready to put (or, more correctly according to database terminology, insert) data about some users into that table. Using Notepad, prepare an `insertUsers.sql` file containing the following SQL code
    a. the two `INSERT` commands shown on pages 136 and 137.
    b. to similarly insert the users shown on page 138, plus
    c. to similarly insert another ten users that you make up – use names of any famous people/movie characters/any others that you choose.
14. Use that SQL source file in the `mysql` client to insert all of those users.
15. Once you have successfully inserted all of the users into the users table, we are ready to try executing some SQL queries that will extract data that we want from the table. Execute the `SQL` queries shown on pages 140 through 165. In these, the author shows you how to look up data in a database using the SELECT command, how to apply criteria, and so on. Since each of these queries is just a single line of code specifying some data to be extracted, and the purpose of this part is for you to gain some practice executing SQL queries, it is okay to type those directly at the command prompt rather than into a source file.
16. Once you have successfully executed all of the queries given in the textbook, stop the logging process by typing `notee` at the command prompt.
17. Exit the `mysql` client.

Part 3: Creating a database for the clothing store

In this third part of the assignment, you will create a database for our online clothing store.

1. Prepare separate `.sql` files to do the tasks listed in parts a) through d) below, with each file's name describing what the code within it does, as you saw in Part 2 earlier.
    a. Create a database named `clothingStore`, set it as the database we are going to use, and create the following tables in it:
        i. `Customers` – containing fields with appropriate data-types and field sizes for
            1. an auto-incrementing ID
            2. first name
            3. last name
            4. street address
            5. city
            6. state
            7. zip

ii. `Products`
    1. auto-incrementing ID
    2. product name (currently, this would be simply “shirt”, “pants” or “shoes” in our case)
    3. description
    4. price
iii. `Orders`
    1. auto-incrementing ID
    2. product ID foreign key referencing the product ID from the Product table
    3. quantity
    4. customer ID foreign key referencing the product ID from the Customer table

b. Add 5 customers whom you make up, to the customer table
c. Add our three generic products of shirt, pants and shoes with descriptions and prices that you make up
d. Add three orders for each customer – one for each of our products – with quantities of each that you make up

2. Once your `SQL` source files are ready, run the `mysql` client using the `runClient` batch file and enter your password when prompted.
3. Start up a logfile named `clothingStore.txt`
4. Use the `.sql` files that you prepared in 1(a) through 1(d) above to create the database and tables and insert data into the tables, correcting any errors within your SQL code in those files whenever you receive error messages.
5. Once you have completed step #4 above successfully, run some SQL queries to show all of the data that you have inserted into the tables.
6. Stop the logfile
7. Finally, copy ALL of the SQL code from the various `.sql` files that you prepared for steps 1(b) through (d) into the file for 1(a), so that you now have a single `.sql` file that contains your code to create the `clothingStore` database, create tables in it, and insert data into those tables. Save as `createClothingStoreDatabase.sql`.

Submitting your files for this assignment:

1. Paste all the screenshots you made earlier into a single Word document.
2. Create a zip file containing
    a. the Word document
    b. your `runClient.bat` file
    c. all of the `.sql` files you have created (for Part 3, include only the final `createClothingStoreDatabase.sql` file – I will run this to try recreating your database on my computer)
    d. all of the logfiles you have created
3. Submit your zip file on Blackboard by the deadline shown there for this assignment.